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Industrial Boiler Safety & Monitoring System (IBSMS)

Project Report

1. Introduction

The Industrial Boiler Safety & Monitoring System (IBSMS) is a simulation-based embedded Linux project designed to monitor critical boiler parameters such as temperature, pressure, water level, and gas leakage.

The system continuously collects sensor data, analyzes it in real time, and generates warnings when unsafe conditions occur. It mimics real industrial safety systems used in power plants and manufacturing units.

The project demonstrates the use of Linux system programming concepts such as processes, threads, inter-process communication (IPC), signals, and file handling.

2. Problem Statement

Industrial boilers work under high temperature and pressure conditions, making manual monitoring difficult and unsafe. Existing systems may fail to detect abnormal conditions quickly and provide continuous monitoring.

The Industrial Boiler Safety & Monitoring System (IBSMS) provides real-time monitoring of boiler parameters, automatically detects unsafe conditions, generates warning alerts, maintains logs, and supports concurrent monitoring using Embedded Linux and IPC mechanisms.

3. Objectives

The main objectives of the project are:

- To simulate real-time boiler monitoring
- To detect unsafe conditions and generate alerts

- To implement multi-process and multi-threaded architecture
- To demonstrate IPC mechanisms like pipe, shared memory, and FIFO
- To implement logging and signal-based shutdown

4. Technologies and Components Used

Technologies Used

Category	Details
Language	C Programming
OS	Linux (Ubuntu)
Concepts	Process Management, Multithreading, IPC, Signal Handling, File Handling, Mutex

Components Used

Component	Function
Controller Process	Analyzes data, generates alerts, maintains logs
Sensor Process	Generates sensor values and sends data
Sensor Threads	Monitor temperature, pressure, water level, and gas

IPC Mechanism

IPC Method	Purpose
Pipe	Transfers sensor data
Shared Memory	Stores latest sensor values
FIFO	Sends warning alerts

5. System Architecture

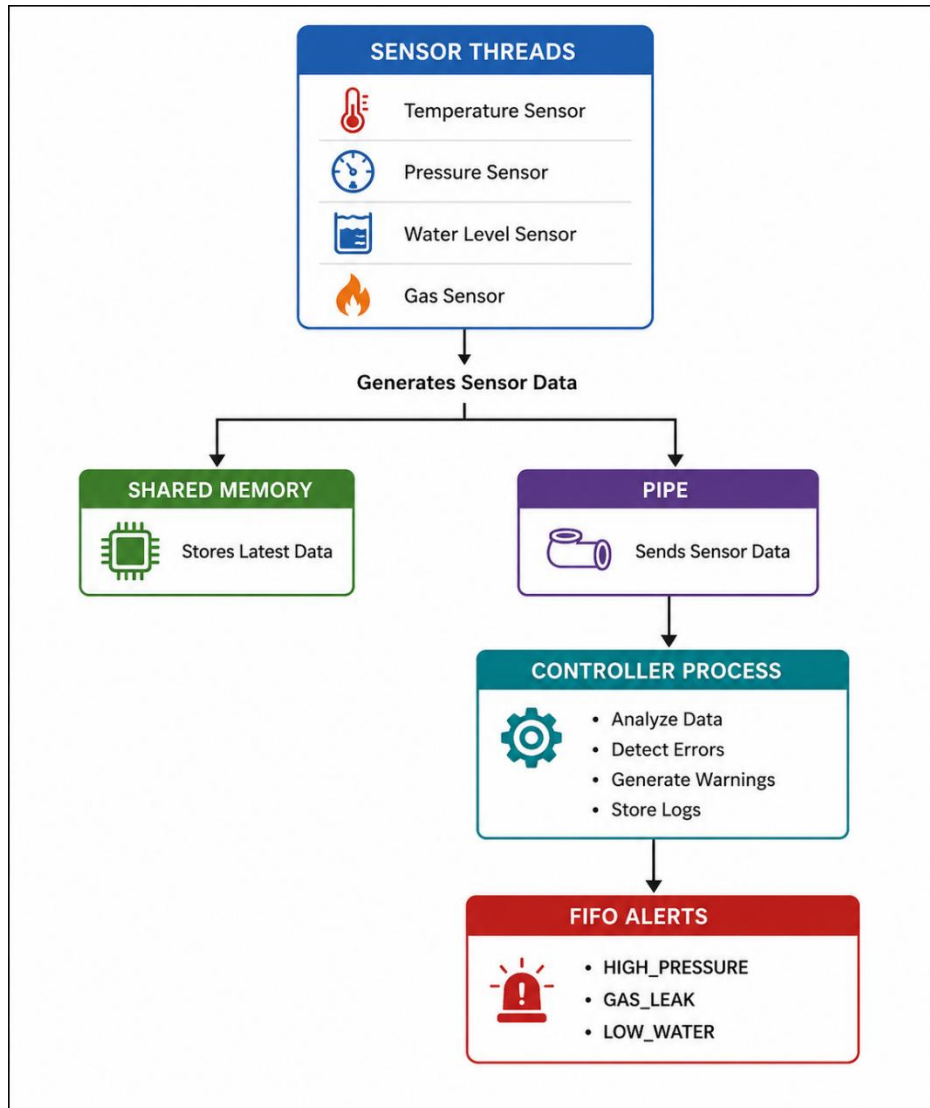


Fig. 1: System Architecture

6. Workflow

- The system initializes IPC mechanisms such as Pipe, Shared Memory, and FIFO.
- The program creates controller and sensor processes using `fork()`.
- The sensor process creates multiple sensor threads for temperature, pressure, water level, and gas monitoring.
- Sensor threads continuously generate sensor values and update shared memory.
- Sensor data is sent to the controller process through the pipe.
- The controller process analyzes the received data and checks for unsafe conditions.

- If abnormal conditions are detected, warning alerts are generated using FIFO and stored in log files.
- The system continues monitoring until a shutdown signal (SIGINT) is received, after which all resources are safely released.

7. Description of All Files

Project Structure

boiler_monitor/

1. src/
2. include/
3. logs/
4. Makefile

File Description

1) main.c

Main execution file of the project.

Functions:

- Initializes the system
- Creates processes
- Starts monitoring

2) controller.c

Implements controller process logic.

Functions:

- Reads sensor data
- Detects unsafe conditions
- Generates alerts

- Displays warnings

3) sensor.c

Implements sensor simulation.

Functions:

- Creates sensor threads
- Generates sensor values
- Updates shared memory
- Sends data through pipe

4) ipc.c

Implements IPC mechanisms.

Functions:

- Pipe creation
- Shared memory setup
- FIFO communication
- IPC cleanup

5) logger.c

Handles logging operations.

Functions:

- Creates log file
- Stores alerts
- Maintains logs

6) Header Files

Contains:

- Function declarations
- Structures
- Constants
- Shared definitions

7) Makefile

Used for project compilation.

8. Advantages

Real-time monitoring simulation

- Fault detection system
- Uses multiple IPC mechanisms
- Modular and scalable design
- Industry-relevant implementation

9. Applications

- Industrial boiler systems
- Power plant monitoring
- Embedded Linux control systems
- Safety-critical environments

10. Conclusion

The IBSMS project successfully demonstrates core Linux system programming concepts in a real-world scenario by integrating process control, threading, IPC, signal handling, and file operations into a cohesive safety monitoring system.